

EDEXCEL - LONDON EXAMINATIONS

PMT

Stewart House 32 Russell Square London WC1B 5DN

June 2001

Advanced Supplementary/Advanced Level

General Certificate of Education

Subject MECHANICS 6677

Paper No. M1

Question number	Scheme	Marks
1.	$3 \rightarrow \quad \leftarrow 2$ Before $0.5 \quad \quad 0.2$ $1.5 \rightarrow v$ After (Mom^m eqn. with 4 terms) (a) $0.5 \times 3 - 0.2 \times 2 = 0.5 \times 1.5 + 0.2 \times v$ $\Rightarrow v = \underline{1.75 \text{ ms}^{-1}}$ (b) $I = 0.2(2 + 1.75)$ $= \underline{0.75 \text{ Ns}}$	m1 A1 A1 (3) m1 A1 ✓ A1 (3) (6)

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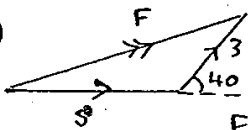
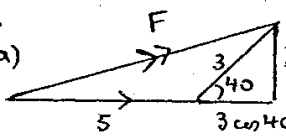
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2.	<u>EITHER</u> (a)  Vector Δ attempt Correct $F^2 = 5^2 + 3^2 - 2 \cdot 5 \cdot 3 \cos 140 \quad (\text{cos rule})$ $\rightarrow F \approx \underline{7.55 \text{ N}}$ (b) $\frac{F}{\sin 140} = \frac{3}{\sin \theta} \Rightarrow \theta \approx \underline{14.8^\circ}$ <u>OR.</u> (a)  Vector Δ attempt correct $F^2 = (5 + 3 \cos 40)^2 + (3 \sin 40)^2$ $F \approx \underline{7.55 \text{ N}}$ (b) $\tan \theta = \frac{3 \sin 40}{5 + 3 \cos 40}, \quad \theta \approx \underline{14.8^\circ}$ <u>OR.</u> (a) $\underline{P} = \begin{pmatrix} 5 \\ 0 \end{pmatrix}$ or $5\hat{i}$ $\underline{Q} = \begin{pmatrix} 3 \cos 40 \\ 3 \sin 40 \end{pmatrix}$ or $3 \cos 40\hat{i} + 3 \sin 40\hat{j}$ $\Rightarrow \underline{F} = \begin{pmatrix} 5 + 3 \cos 40 \\ 3 \sin 40 \end{pmatrix}$ $ \underline{F} = \sqrt{(5 + 3 \cos 40)^2 + (3 \sin 40)^2}$ $\approx \underline{7.55 \text{ N}}$ (b) $\tan \theta = \frac{3 \sin 40}{5 + 3 \cos 40}$ $\approx \underline{14.8^\circ}$	m1 A1 m1 A1 A1 (5) m1 A1, A1 (3) (8) m1 A1 m1 A1 ✓ A1 (5) m1 A1, ✓ A1 (3) (8) m1 A1 m1 A1 ✓ A1 (5) m1 A1 ✓ A1 (3) (8)

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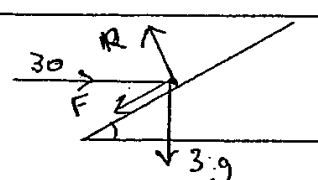
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3.	(a) Distance = $\frac{1}{2} \times (30+17) \times 3 + 4 \times 17$ $= 138.5 \text{ m.}$ <u>OR</u> $\frac{1}{2} \times 3 \times (30-17) + 3 \times 17 + 4 \times 17$ $= 138.5 \text{ m}$ (b) Str. line graph $\Rightarrow$ const. decel² $"F=ma" \Rightarrow F \text{ const}$ (c) Decel² = $\frac{30-17}{3}$ Force = $1200 \times \left(\frac{30-17}{3}\right) = 5200 \text{ N}$	m1 A1, m1 A1 (4) m1 A1, m1 A1 m1 A1 c.s.o. (2) m1 m1 A1 (3) (9)
4.	(a)  Diag. with 4 forces marked (Allow F & R combined if clear) (b) R(↑) $R = 3g \cos 30^\circ + 30 \sin 30^\circ$ (3 terms) $= 40.46 \dots \approx 40.5 \text{ or } 40 \text{ N.}$ (c) R(←) $F = 30 \cos 30^\circ - 3g \sin 30^\circ$ (3 terms) $F = \mu R, \Rightarrow \mu = \frac{F}{R} = \frac{11.28}{40.46}$ $\approx 0.28 \text{ (or } 0.279)$	B2 -1 e.e. (2) m1 A2 -1 e.e. A1 (4) m1 A1 m1, m1 A1 (5) (11)

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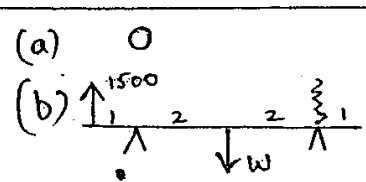
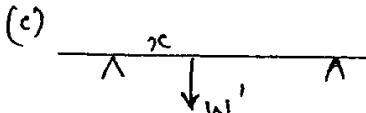
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5.	(a)  (b) $M(D): 2W = 1500.5$ $\Rightarrow W = 3750 \text{ N}$ [If moments about another pt: m1 for a complete method to get W, A1 for a moments eqn² correct.] (c)  $M(D) \quad 1500.5 = W'(4-x)$ $M(C) \quad 1000.5 = W'x$ Solve $\rightarrow W' = 3125 \text{ N}$ (d) $x = 1.6 \text{ m}$ (e) AB remains straight line (o.e.)	B1 (1) m1 A1 A1 (3) m1 A1 m1 A1 (m1) A1 (6) m1 A1 (2) B1 (1) (13)

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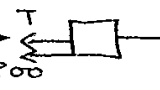
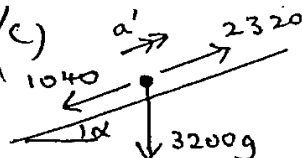
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6.	(a) Car + Van: $3200a = 2320 - 800 - 240$ $a = 0.4 \text{ ms}^{-2}$ (b) Car:  $1200a = T - 240$ $\rightarrow T = 720 \text{ N}$ [OR Van:  $2000a = 2320 - 800 - T$ $\rightarrow T = 720$ NB If use eqn^2 for car & van alone, allow M1 A2 for one eqn^2 involving T, then M1 A1 for a second eqn^2 <u>provided</u> it is part of a complete method to find a/T. Then A1 A1 for a & T. (c)  $3200a' = 2320 - 1040 - 3200g \cdot \frac{1}{20}$ (4 terms) $a' = -0.09 \text{ ms}^{-2}$ $\Rightarrow$ magn. 0.09 ms^{-2} speed decreasing	M1 A1 A1 (3) M1 A2 ✓ -1 e.e. A1 (4) M1 A2 -1 e.e. M1 A1 A1 (6) (13)

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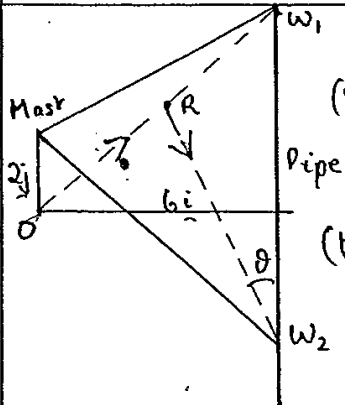
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7.	 (a) $\underline{w}_1 = 2\underline{j} + 6\underline{i} + 6\underline{j}$ $= 6\underline{i} + 8\underline{j}$ (b) $OW_1 = \sqrt{(6^2 + 8^2)} = 10 \text{ km}$ $\Sigma \text{sr}^2 \text{ time} = \frac{10}{5} = 2 \text{ hrs}$ (c) $\underline{w}_2 = 2\underline{j} + 6\underline{i} - 6\underline{j}$ $= 6\underline{i} - 4\underline{j}$ (d) P.v. of rescue party after 1 hour = $\underline{R} = 3\underline{i} + 4\underline{j}$ $\underline{R}\underline{w}_2 = 3\underline{i} - 8\underline{j}$ $\tan \theta = \frac{3}{8} = 20.6^\circ$ $\Rightarrow \text{Required bearing} = 180^\circ - 20.6^\circ$ $= 159.4^\circ$	BI BI (2) MI (3) BI, MI AI (3) MI AI MI AI MI MI AI (7) (15)